Extra Bites

A strategically selective Amateur Extra study system

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What This Is

Extra Bites is not a class. It is a coherent two-year set of short club-meeting concept lessons, a few longer concept sessions, and printable self-study supplements. The whole system is built on **strategic scoping**: it deliberately does not try to teach every question in the 599-question Extra pool. It targets the smallest realistic subset of the pool that still gives a motivated General-class operator a pass-grade path.

The Three Layers

Layer	Length	Job	Where it lives
Extra Course	30-45 min	Owns the hardest included material that cannot be honestly taught in 10 minutes. Map-style sessions with diagrams and worked examples.	Twice per year at a regular club meeting program slot.
Extra Bite	5-10 min	One compact concept that unlocks several related exam questions. Plain language, one diagram, one example, exam IDs.	Short segment at any monthly club meeting. Handout goes home.
Extra Dessert	self-study	Easier or memorization-heavy material: rules, privileges, operating procedures, propagation tables, recognition drills. No live teaching needed.	Distributed alongside Bites and at meetings. Done at home.

Why Strategic Scoping

The Extra exam has 50 questions drawn one per exam group from a 599-question pool. Passing requires 37 correct. Two facts follow:

- **Exam coverage, not pool coverage, is the planning target.** Setting aside 12 questions from one pool group does not remove 12 exam questions; it removes about one. The right unit of scoping is the **exam group**, not the raw pool question.
- **Time is finite.** A two-year club program cannot honestly teach every group in 5-10 minute chunks. Spending equal time on a math-heavy abstract group and a high-leverage pattern-recognition group wastes the easier wins.

The system **reserves 7 of the 50 exam groups for separate self-study** (about 84 pool questions) and uses the recovered capacity to teach the remaining 43 well.

Pass Math

Quantity	Working assumption
Exam Qs on included material	about 43 of 50
Exam Qs on excluded material	about 7 of 50
Accuracy on covered	about 85%
Guess rate, uncovered	about 25%
Expected score	43 × 0.85 + 7 × 0.25 ≈ **38.3**
Pass threshold	37 of 50

Thin but real margin of one question above pass. Buffer comes from learners who exceed 85% on covered material – realistic when the teaching is concept-first.

Reserved For Separate Self-Study

7 groups, 84 pool questions, 7 of 50 exam questions. Real EE topics; just hard to deliver in 10-min segments.

Grp	Topic	Qs
E5C	Complex impedance, phasors	12
E5B	Phase angles, time constants, susceptance	12
E7G	Operational amplifiers	12
E9G	Smith chart	11
E8C	Digital mode concepts, QAM	15
E8D	DSP, coding, signal-processing	11
E6C	Digital/logic component behavior	11

How To Use The System: Modular Compression

The 22-meeting program length is the **gentlest** setting, not a fixed duration. Every Bite, Course, and Dessert is an independent module, so a club or group dials the pace to fit – same materials, same handouts, same recordings, only the schedule changes.

Pace	Bites / mtg	How it runs	Finishes in
Standard	1	One short Bite per monthly meeting; Courses in their scheduling windows. Relaxed; never crowds club business.	22 months
Accelerated Club	2	Two Bites per monthly meeting – only 10-15 min of program time; Courses standalone.	13 meetings

Study-Group Sprint	3, weekly	Three Bites per session, run weekly; Courses standalone. Still under an hour per session.	10 weeks

The four Courses always run as their own full-length session; that is built into every count above. The Bites are not mainly a cram tool – their real value is making the next license feel reachable and putting the material in front of a ham early, so a later upgrade is familiar ground instead of first contact. An individual can self-study the same modules at any pace, and VEs or Elmers can hand the whole bundle to an upgrade candidate as a prep packet.

Success Definition

After two years a regular attendee should be able to say: *"I have seen this material. I know what kind of idea each question is asking about. With focused review of the included groups, passing feels reachable."* The system does not promise mastery. It promises a defensible pass-grade path with the least intimidating possible on-ramp.

System Map

Every Course, Bite, and Dessert with its primary exam group and question count

4 Extra Courses (hardest included clusters, 30-45 min each)

ID	Title	Primary groups	Pool Qs
EC-A	Resonance, Q, and bandwidth	E5A	13
EC-B	Receiver measurement: VNA, S-parameters, noise floor, MDS	E4B + E4C	25
EC-C	Antenna math and feed lines: dB, ERP/EIRP, velocity factor	E9A + E9F	24
EC-D	Modulation, digital emissions, modulation index	E8A + E8B	22
Total		**7 groups**	**84**

Teaching Sequence: 18 Bites Interleaved With 4 Courses

Bites are grouped four at a time. Standalone Bites scheduled first within each group of four; Course-dependent Bites last. Each Course gets a **scheduling window** instead of a fixed slot. Only two Bites have Course dependencies: B13 needs EC-B (window: before Bite 13), B17 needs EC-A (window: before Bite 17). EC-C and EC-D have no Bite dependencies.

Slot	Title	Group	Qs
B1	Power supplies and regulators	E7D	15
B2	Practical wire antennas, phased arrays, and ground effects	E9C	14
B3	Semiconductor device recognition	E6A	12
B4	MMICs and RF packaging	E6E	12
EC-A	**Course: Resonance and Q (E5A) — window: any time before Bite 17 — 13 Qs**		
B5	Skin effect, reactive power, RF parasitics	E5D	12
B6	Amplifier classes and transistor stages	E7B	12
B7	Software defined radio and DSP basics	E7F	14
B8	Oscillators, synthesizers, PLL	E7H	13
EC-B	**Course: Receiver measurement (E4B + E4C) — window: any time before Bite 13 — 25 Qs**		
B9	Yagis, parabolic dishes, short antennas	E9D	12
B10	Diodes	E6B	11
B11	Inductors, transformers, piezoelectric devices	E6D	11
B12	Electro-optical devices and optoelectronics	E6F	11
B13	Receiver dynamic range and intermod (uses EC-B)	E4D	13
EC-C	**Course: Antenna basics and feed lines (E9A + E9F) — no Bite deps, schedule anytime — 24 Qs**		
B14	Digital logic circuits and truth tables	E7A	11
B15	Modulators, detectors, and mixers	E7E	11
B16	Antenna patterns and modeling	E9B	11
B17	Filters and matching networks (uses EC-A)	E7C	11
EC-D	**Course: Waveforms, RMS, PEP + modulation (E8A + E8B) — no Bite deps, schedule anytime — 22 Qs**		
B18	Impedance matching and phasing lines	E9E	10
Total	**Bites + Courses**	**25 groups**	**300**

22 Extra Dessert Supplements (self-study, ordered by impact)

ID	Title	Group / role	Qs
D1	Specialized propagation (EME, meteor scatter, auroral, microwave)	E3A	14
D2	Noise, interference, and mitigation	E4E	14
D3	HF digital modes	E2E	13
D4	Propagation effects (transequatorial, long-path, sporadic-E)	E3B	13
D5	Safety, RF exposure, and grounding	E0A	12
D6	Automatic and remote control, foreign operations, emission rules	E1C	12
D7	Amateur space and Earth stations, telemetry, balloons	E1D	12
D8	Satellite operations	E2A	12
D9	Television practices (ATV, SSTV)	E2B	12
D10	DX, contest, and remote operating	E2C	12
D11	Propagation prediction and space weather	E3C	12
D12	Frequency privileges and band edges	E1A	11
D13	Station restrictions, RACES, special operations	E1B	11
D14	VE and exam administration	E1E	11
D15	Miscellaneous rules: external amps, spread spectrum, STA	E1F	11
D16	VHF/UHF digital, APRS, EME, meteor scatter procedures	E2D	11
D17	Test equipment: scopes, spectrum analyzers, antenna analyzers	E4A	11
D18	Receiving antennas: RDF, Beverages, loops	E9H	11
D19	Exam blueprint and score tracker	system organizer	—
D20	Device and circuit symbol drill	supports EC-B, all E6/E7 Bites	—
D21	Antenna figure and pattern drill	supports EC-C, B2, B9, B16, B18	—
D22	Final mixed practice and weak-area diagnostic	system organizer	—
Total		**18 groups + 4 drill**	**213**

System Totals

Layer	Groups	Pool Qs
Courses	7	84
Bites	18	216
Desserts (group-owned)	18	213
Drill / organizer Desserts	—	—
Reserved for separate study	7	84
Pool covered	**43 of 50**	**513 of 599**

Study Groups and Self-Study

Running Extra Bites at the Study-Group Sprint end of the compression dial

How A Study-Group Sprint Works

A study group is **not a class** – it is three or four people working through the material together, with no teacher. A weekly Sprint session is short and runs itself: play the three Bite recordings for the week (about 15-30 minutes – the recordings do the teaching), then work that week's Dessert sheets together – compare practice-question answers, talk through whatever was confusing, and quiz each other. The whole session stays comfortably under an hour.

Two things keep the format working: **no one has to teach** – the recordings handle that – and **everyone does their own practice-question drill between sessions**, because the group amplifies self-study, it does not replace it. Four operators working the same material together will catch traps and explain things to each other in ways a single teacher rarely can.

What The System Will Not Do

- It will not lecture every question in the pool.
- It will not teach the 7 groups reserved for separate self-study. Learners who want to master those topics should put time into the recommended outside resources; on the exam itself, anything not covered by the time the exam is taken can be answered by educated guess.
- It will not replace practice questions. Concept exposure plus repeated practice-question drill is the working combination.

Success Target

Move a motivated General-class operator from roughly 25 of 50 on a cold practice exam to roughly 38-40 of 50 on a focused exam, where the 5-7 missed questions are concentrated in the deliberately excluded groups. That is the working pass-grade design.